**Вычисление определенного интеграла методом парабол (методом Симпсона)**

$$\int_a^b f(x)dx \approx$$

$$\frac{h}{3} * \left[ y_0 + 4*(y_1 + y_3 \ldots + y_{2n-1}) + 2*(y_2 + y_4 \ldots + y_{2n-2}) + y_{2n} \right]$$

**где** **h =** $\frac{b-a}{2n}$

$x_k = a + k*h$

$y = f(x_k)$

$(k = 0, 1, \ldots, 2n)$

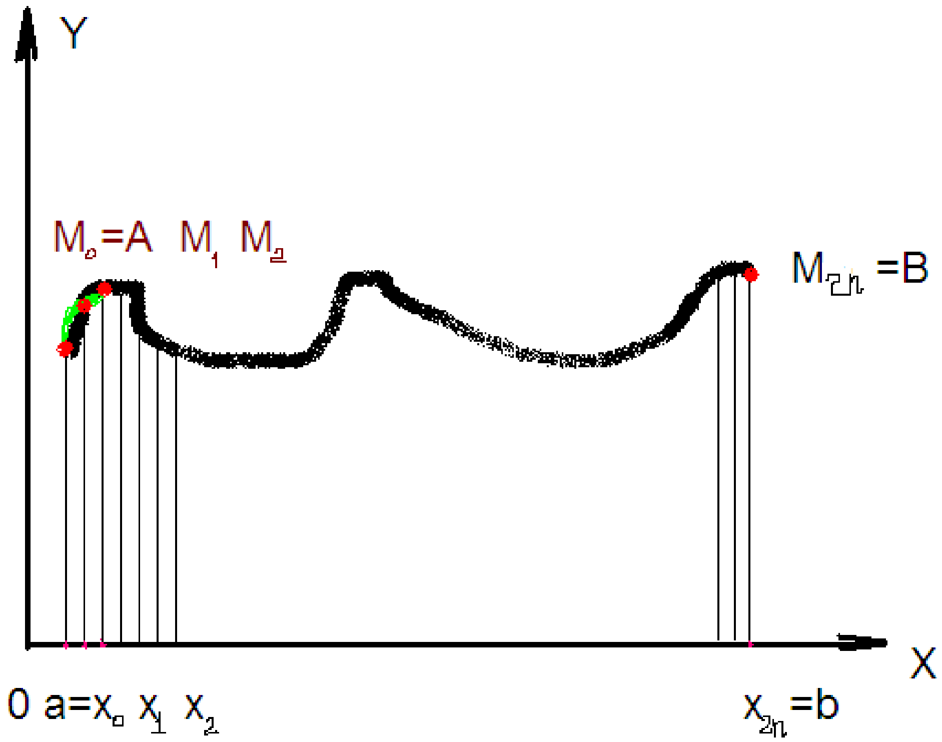


Рис. 1

# МЕТОД ПАРАБОЛ

```
Sub Main()

Dim x, a, b, h, s1, s2, i, i1, i2, f, s As Decimal
Dim n As Integer
        a = 0
        b = 1
        n = 5000
        s = 0
        s1 = 0
        s2 = 0
        h = (b - a) / n
        x = a
        Call func(x, f)
        s = f
        x = b
        Call func(x, f)
        s = s + f
        For x = a + h To b - h Step 2 * h
            s1 = s1 + Math.Sin(x) / x
        Next x
        i1 = 4 * s1
        For x = a + 2 * h To b - 2 * h Step 2 * h
            s2 = s2 + Math.Sin(x) / x
        Next x
        i2 = 2 * s2
        i = h * (i1 + i2 + s) / 3
        Console.WriteLine("Интеграл равен i= {0}", i)
        Stop
   End Sub

    Sub func(ByVal d, ByVal f)
        f = Math.Sin(d) / d
    End Sub
```